

RESPONSE TO OSCILLATING INPUT

$$\frac{dy}{dt} = ay + \cos \omega t$$

$$y(t) = \text{"oscillating response"}$$

$$\underbrace{\left(\frac{dy}{dt} \right)}_{\text{(Growth term)}} \propto \underbrace{y}_{\text{(Balance)}}$$

(a)

Interest added on to y

 $(\cos \omega t)$

→ Source term, forcing term, input

→ Alternatively, e^{st}

exponential term

grows or decays with the solution

→ $\cos \omega t$ → oscillating term

→ Examples/Applications

1. AC circuits
2. Vibrations
3. Applications involving circular motion

We have:

Initial condition or starting deposit

$$y = y(0) \text{ at } t = 0$$

Solution:

$$\underbrace{(\text{Complete solution})}_y = \underbrace{(\text{Particular solution})}_{y_p} + \underbrace{(\text{Homogenous solution})}_{y_h}$$

Let's start with the particular solution:

(y_p)

Consider these points:

- ① We are looking for a particular solution like $y_p(t) = M \cos \omega t$
- ② Recall that for e^{st} forcing term, we chose $y_p(t) = Y e^{st}$
- ③ However, the above approach won't work.
- ④ Our y_p should take into consideration, the forcing term and its derivative

Clue: The derivative of a cosine is a sine function. So, sines are going to get in there and we have to allow them into the particular solution

So we look for:

$$y_p(t) = M \cos \omega t + N \sin \omega t$$

{where M, N are constants}

③

Find:

M & N in equation ③ by substituting y_p into equation ①

Aside:

- ① There are 3 ways to answer "oscillating response" problem
- ② Eq ③ is the most straight forward, but not the best!

(44)

KVB

$$\therefore \frac{dy_p}{dt} = ay_p + \cos \omega t$$

Substituting $y_p = M \cos \omega t + N \sin \omega t$, we have,

$$\frac{d}{dt}(M \cos \omega t + N \sin \omega t) = a(M \cos \omega t + N \sin \omega t) + \cos \omega t$$

$$\Rightarrow -\omega M \sin \omega t + \omega N \cos \omega t = aM \cos \omega t + aN \sin \omega t + \cos \omega t$$

Re-arranging the terms

$$\begin{aligned} \Rightarrow (-\omega M \sin \omega t - aN \sin \omega t) &= 0 + \cos \omega t \\ (\omega N \cos \omega t - aM \cos \omega t) & \end{aligned}$$

Matching the sine & cosine term, we have

$$\boxed{\cos \omega t} \quad \omega N \cos \omega t - aM \cos \omega t = \cos \omega t$$

$$\Rightarrow \boxed{-aM + \omega N = 1} \quad \rightarrow (4a)$$

$$\boxed{\sin \omega t} \quad -\omega M \sin \omega t - aN \sin \omega t = 0$$

$$\Rightarrow \boxed{-\omega M - aN = 0} \quad \rightarrow (4b)$$

$\therefore$ Equations (4a) & (4b) are; two equations with two unknowns, (M, N) .

$\therefore$ Solve for M & N

Solve for M, N : (In Matrix form.)
 $Ax = B$

$$\begin{bmatrix} -w & -a \\ -a & w \end{bmatrix} \begin{bmatrix} M \\ N \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\underbrace{\begin{bmatrix} -w & -a \\ -a & w \end{bmatrix}}_A \quad \underbrace{\begin{bmatrix} M \\ N \end{bmatrix}}_x \quad \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_B$

Recall:
 $x = A^{-1}B$

A^{-1} : $\frac{\text{adjoint}(A)}{\det(A)} \Rightarrow \frac{\begin{bmatrix} w & -(-a) \\ -(-a) & -w \end{bmatrix}}{(-w)(w) - (-a)(-a)}$

$$\Rightarrow \frac{\begin{bmatrix} w & a \\ a & -w \end{bmatrix}}{-w^2 - a^2} \quad \therefore A^{-1} = \begin{bmatrix} \frac{w}{-w^2 - a^2} & \frac{a}{-w^2 - a^2} \\ \frac{a}{-w^2 - a^2} & \frac{-w}{-w^2 - a^2} \end{bmatrix}$$

Now: $\begin{bmatrix} M \\ N \end{bmatrix} = \begin{bmatrix} \frac{w}{-w^2 - a^2} & \frac{a}{-w^2 - a^2} \\ \frac{a}{-w^2 - a^2} & \frac{-w}{-w^2 - a^2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} M \\ N \end{bmatrix} = \begin{bmatrix} \frac{a}{-w^2 - a^2} \\ \frac{-w}{-w^2 - a^2} \end{bmatrix} = \begin{bmatrix} \frac{-a}{w^2 + a^2} \\ \frac{w}{w^2 + a^2} \end{bmatrix}$$

$$\Rightarrow \boxed{\begin{array}{l} M = \frac{-a}{w^2 + a^2} \rightarrow (5a) \\ N = \frac{w}{w^2 + a^2} \rightarrow (5b) \end{array}}$$

Substitute (5a) & (5b) into equation (5)

$$y_p = M \cos \omega t + N \sin \omega t$$

$$\Rightarrow y_p = \left(\frac{-a}{\omega^2 + a^2} \right) \cos \omega t + \left(\frac{\omega}{\omega^2 + a^2} \right) \sin \omega t$$

→ Particular solution

→ Oscillation that keeps going

→ If we are listening to the radio
or

→ If we have alternating current

→ y_p is the solution your ears listen to!

Now, moving on to the Null or Homogenous Solution.

Substitute y_h in equation (1) (6)
without the source term.

$$\frac{dy_h}{dt} = ay_h$$

$$\Rightarrow \frac{dy_h}{y_h} = a dt$$

$$\Rightarrow \ln y_h = at + C_1$$

$$y_h = C e^{at}$$

$$\Rightarrow y_h = e^{at} e^{C_1}$$

→ Homogenous solution

→ If a is negative or negligible this term disappears.

SECOND FORM OF THE FULL SOLUTION

Let's take another look at ' y_p ':

I believe that $y_p = M \cos \omega t + N \sin \omega t$
can be written in a different way

- Cosines & sines are out-of-phase
- combining cosines & sines would be nice!
- We can think of writing y_p as " $\cos(\omega t - \alpha)$ "

So, we look for:

$$y_p = G \cos(\omega t - \alpha)$$

Gain
or
Magnitude

Angle:
Represents
Phase-
Shift

(12)
Polar
Co-ordinates
Combines
Sines &
Cosines

Let's match ' y_p ' from equation (3) to ' y_p ' from (12):

Remember: We know
our M, N :

$$M = \frac{-a}{\omega^2 + a^2}, N = \frac{\omega}{\omega^2 + a^2}$$

$$[y_p = M \cos \omega t + N \sin \omega t] \text{ to } [y_p = G \cos(\omega t - \alpha)]$$

$$\begin{aligned} \therefore M \cos \omega t + N \sin \omega t &= G \cos(\omega t - \alpha) \\ &= G(\cos \omega t \cos \alpha + \sin \omega t \sin \alpha) \\ &= G \cos \alpha \cos \omega t + G \sin \alpha \sin \omega t \end{aligned}$$

⇒

$$\begin{aligned} M &= G \cos \alpha \\ N &= G \sin \alpha \end{aligned}$$

(13)

Find:
 G, α

TRICK: To find G, α use $\sin^2 \alpha + \cos^2 \alpha = 1$

From (13) let's do some trigonometry

$$M^2 + N^2 = G^2 (\cos^2 \alpha + \sin^2 \alpha)$$

1

$$\Rightarrow G = \sqrt{M^2 + N^2} \quad \rightarrow (14)$$

Again from (13)

$$\frac{G \sin \alpha}{G \cos \alpha} = \frac{N}{M} \Rightarrow \tan \alpha = \frac{N}{M}$$

(15)

Full solution:

$$y = y_p + y_h$$

Combine (12) & (8)

$$y(t) = G \cos(\omega t - \alpha) + C e^{at}$$

$\rightarrow (16)$

To evaluate C , in equation (16), we use the

initial condition:

$$y = y(0) \text{ at } t=0 \quad \text{Equation (2)}$$

Substitute initial condition in equation (16)

$$y(0) = G \cos(\omega \times 0 - \alpha) + C e^{a \times 0}$$

1

$$= G \cos(-\alpha) + C$$

$\rightarrow (17)$

$$\therefore y(0) = G \cos(\alpha) + C \Rightarrow C = y(0) - G \cos(\alpha)$$

Combine (16) & (17)

$$y(t) = G \cos(\omega t - \alpha) + [y(0) - G \cos(\alpha)] e^{at}$$

$$\left\{ \begin{array}{l} \text{where } G = \sqrt{M^2 + N^2} \\ \alpha = \tan^{-1} \frac{N}{M} \end{array} \right.$$

→ This is $y(t)$ in terms of

1. Magnitude or Gain, and
2. Phase

RESPONSE TO ANY INPUT

Solve a first-order ODE for any input:

(with constant, a)

interest rate

source term,
forcing function

$$\frac{dy}{dt} = ay + q(t)$$

⊕ We'll talk this through
the "money-in-the-bank"
analogy

Growth
of
Money

$$\left(\frac{dy}{dt}\right)$$

Interest added
new deposits

Balance
of
Money

$y \rightarrow$ Money-in-the-bank

Start at $y(0)$

$$y(0)$$

$\rightarrow$ Once-in-lifetime
deposit
 $\rightarrow$ Initial deposit

Big formula for first-order linear ODEs

$$y(t) = y(0)e^{at} + \int_{s=0}^{s=t} e^{a(t-s)} q(s) ds$$

Some form
of this term:

Initial deposit
that grows with the
interest rate as an
exponent.

NULL SOLUTION

$\rightarrow$ Matches the deposit
 $\rightarrow$ Each deposit is made
continuously at time t
 $\rightarrow$ Grows over the
remaining time

PARTICULAR SOLUTION

$$y(t) = y(0)e^{at} + \int_{s=0}^{s=t} e^{a(t-s)} q(s) ds$$

Particular solution

- Grows out of the source term
- We make deposits $q(s)$ at any time (s)
- Between $s=0$ & $s=t$
i.e., $\int_{s=0}^{s=t}$
- 's' is the running clock
- 't' is the time at which we look into our account

- $q(s)$ grows in the remaining time between s & t
- Growth is at the rate of $e^{a(t-s)}$

- Adding up deposits and its growth
 $\int_{s=0}^{s=t} e^{a(t-s)} q(s) ds$
in continuous time.

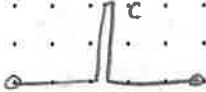
EXAMPLES OF $q(t)$: { where we didn't have this formula, but we got to the solution directly }

$q(t) = C$ [CONSTANT]

$= e^{st}$ [EXPONENTIAL]

$= \cos \omega t$ [OSCILLATING (OR SINUSOID)]

$=$ STEP FUNCTION  [LEFT SWITCH ON/OFF]

$=$ DELTA FUNCTION  [IMPULSE]

$= t, t^2, t^3, \dots$

$=$ COMBINATIONS OF ABOVE

EXAMPLE : $q(t) = C$ [constant]

$$y(t) = y(0)e^{at} + \int_{s=0}^{s=t} e^{a(t-s)} q(s) ds$$

→ Complete Solution

$$\int_{s=0}^{s=t} e^{a(t-s)} q(s) ds = \int_{s=0}^{s=t} e^{at} e^{-as} C ds$$

$$= Ce^{at} \int_{s=0}^{s=t} e^{-as} ds$$

$$= Ce^{at} \left[-\frac{1}{a} e^{-as} \right]_{s=0}^{s=t}$$

$$= Ce^{at} \left[-\frac{1}{a} e^{-at} - \left(-\frac{1}{a} e^{-a(0)} \right) \right]$$

$$= Ce^{at} \left[-\frac{1}{a} e^{-at} + \frac{1}{a} \right]$$

$$= \frac{C}{a} [e^{at} - e^{at} e^{-at}]$$

$$= \frac{C}{a} [e^{at} - 1]$$

$$\therefore y(t) = y(0)e^{at} + \frac{C}{a} [e^{at} - 1]$$

EXAMPLE: $q(t) = e^{st}$

$$y(t) = y(0)e^{at} + \int_{s=0}^{s=t} e^{a(t-s)} q(s) ds$$

$$\int_{s=0}^{s=t} e^{a(t-s)} e^{st} ds = e^{at} \int_{s=0}^{s=t} e^{-as} e^{st} ds$$

$$= e^{at} \int_{s=0}^{s=t} e^{s(t-a)} ds$$

Let $t=a$ $\rightarrow$ case of resonance

$$= e^{at} \int_{s=0}^{s=t} e^{s(a-a)} ds$$

$$= e^{at} \int_{s=0}^{s=t} ds$$

$$= e^{at} [s]_0^t$$

$$= te^{at}$$

$$y(t) = y(0)e^{at} + te^{at}$$

SOLVE $\frac{dy}{dx} + a(x)y = q(x)$ by an
INTEGRATING FACTOR $p(x)$

Defⁿ

An integrating factor for a differential equation is a function, $p(x)$, such that

↳ the multiplication by $p(x)$ on each side of the differential equation yields a recognizable derivative on each side

$a(x)$ → Can be a constant
→ Can be dependent on x

METHOD FOR SOLVING FIRST ORDER ORDINARY

DIFFERENTIAL EQUATIONS:

$$\frac{dy}{dx} + a(x)y = q(x)$$

STEP I

CALCULATE THE INTEGRATING FACTOR

$$p(x) = e^{\int a(x) dx}$$

STEP II

MULTIPLY THE FIRST-ORDER ODE by $p(x)$

$$p(x) \frac{dy}{dx} + p(x)a(x)y = p(x)q(x)$$

STEP III

Recognize that left-hand-side is a derivative of a product

$$\frac{d}{dx}(p(x)y) = p(x)q(x)$$

STEP IV

INTEGRATE & SOLVE → $\int d(p(x)y) = \int p(x)q(x) dx + C$